



University of Al-Ameed / College of Medicine
Department of Surgery



WELL, COME & INTRODUCTION TO SURGERY

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Definition

Surgery is the branch of medicine that deals with the physical manipulation of a bodily structure to:

Diagnose,

Prevent,

or Cure a disease.

Such procedures done by a *surgeon*.

Surgical Team



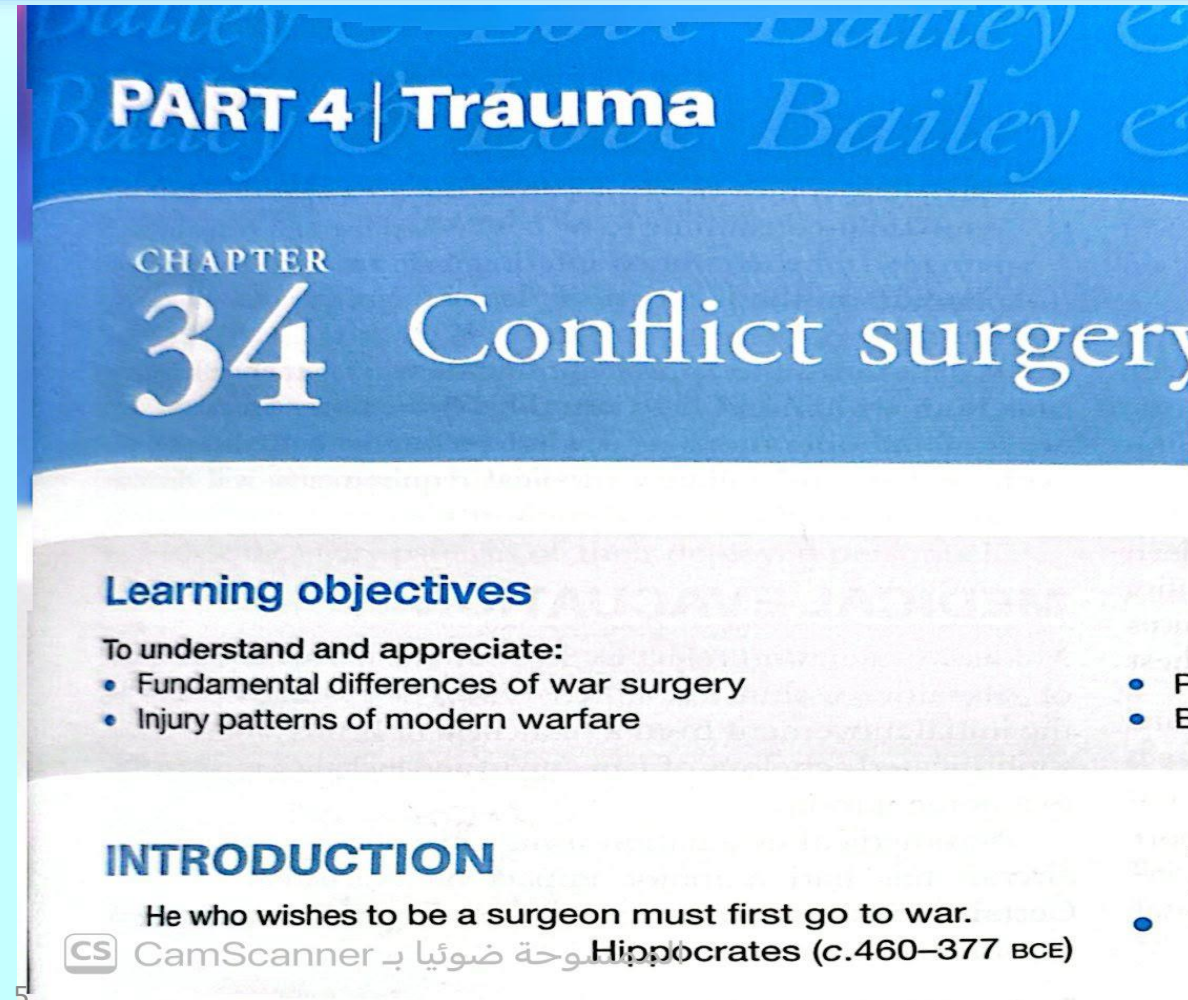
3 Major Obstacles in Surgery:

- Pain
- Bleeding
- Infection

Until the industrial revolution, surgeons were incapable of overcoming these obstacles.

Advances in these fields have transformed surgery from a risky "art" into a scientific plan of treating many diseases & conditions.

History of Surgery



History of Surgery

- **First surgical techniques developed to treat injuries & traumas.**
- **It involved suturing lacerations, amputating unsalvageable limbs, & draining of pus.**
- **Oldest surgery (burr holes or craniotomy).**
- **Intramedullary bones fixation with wooden splints is reported in early Spanish texts.**
- **Anaesthesia by using alcohol or opium is one of the oldest techniques.**

Types of Surgery

- According to ***PURPOSE.***
- According to degree of ***URGENCY.***
- According to degree of ***RISK.***
- According to division of ***SURGERY.***

According to *PURPOSE*

1. Diagnostic.
2. Exploratory.
3. Curative.
4. Palliative.
5. Transplant.

According to Degree of RISK (Magnitude, Extent)

- ❖ Minor
- ❖ Medium
- ❖ Major
- ❖ Super - major

According to Degree of *URGENCY*

- **Emergency** (without delay i.e. immediate surgery)
- **Urgent** (within 1-2 days).
- **Planned (Elective)**

According to division of ***SURGERY***

- General surgery.
- Paediatric surgery.
- Cardiothoracic & Vascular surgery.
- Plastic or reconstructive surgery.
- Maxillofacial surgery.
- Gastro-enteric surgery.
- Neurosurgery.
- Urology.

Courses of Surgical Care

- Pre - operative.
- Per- operative.
- Post - operative.

1- Pre-operative care

For the patient:

- Medical examination.
- Pre-operative tests.
- Patient signs a consent.
- If expected significant blood loss, so prepare blood.

- **Patients are also instructed to abstain from food or drink (an NPO order 6-8 hours before the procedure), to minimize the effect of stomach contents to reduce the risk of aspiration if the patient vomits during or after the procedure.**
- **In the pre-operative holding area, the patient changes out of his street clothes.**
- **Vital signs are recorded.**
- **Peripheral IV line is placed, and pre-operative medications (antibiotics, sedatives, etc.) are given.**

For the surgical team:

- **Surgical masks are worn by the surgical team to avoid germs on droplets of liquid from their mouths & noses from contaminating the operative site.**
- **The surgeons' & assistants' hands, wrists & forearms are washed thoroughly for at least 4 minutes to prevent germs getting into the operative field, then sterile gloves are placed onto their hands.**

➤ Anesthesia is administered to prevent pain from incision, tissue manipulation & suturing.

➤ Based on the procedure, anesthesia may be:

General anesthesia or Spinal or Regional or Local anesthesia.

➤ With local & spinal anesthesia, the surgical site is anesthetized, but the patient can remain conscious or minimally sedated while general anesthesia renders the patient unconscious.

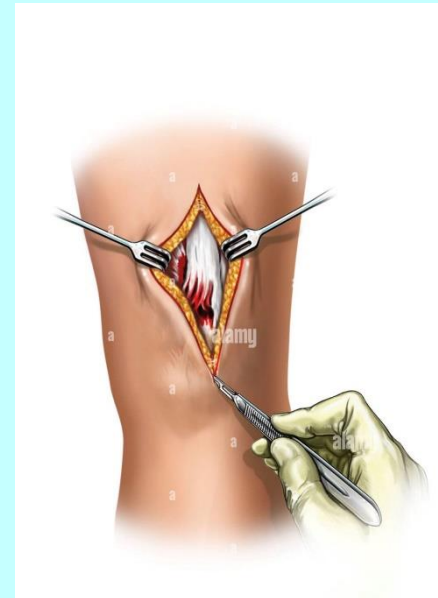
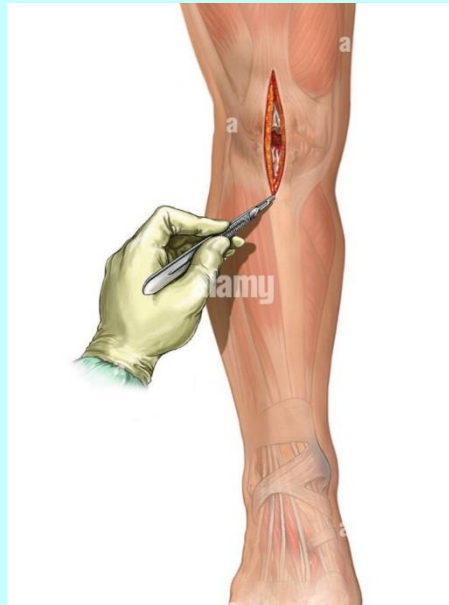
- The skin surface to be operated on, called the *operating field*,
- If hair is present at the *operating field*, , it is clipped off.
- *Operating field* is cleaned & prepared by applying an antiseptic such as povidone-iodine to reduce the possibility of infection.
- Sterile drapes are used to cover the surgical site.

Operating Field



2- Per-operative

- An incision is made to access the surgical site.
- Blood vessels may be clamped or cauterized to prevent bleeding.
- Retractors may be used to expose the site & keep incision open.



Examples of surgical procedures:

- * Haemostasis:* process to prevent & stop bleeding (e.g. direct pressure, cautery, clips, suturing etc.)
- * Ligation:* tying off blood vessels, ducts, or "tubes".
- * Excision:* removal of tissue from the body by a sharp knife, laser, or other cutting tool).
- * Debridement:* Removal of dead or damaged tissue & foreign material from a wound.

* Removal of diseased organs:

- *Appendectomy: an operation to remove the vermiform appendix.*
- *Cholecystectomy: an operation to remove the gallbladder.*

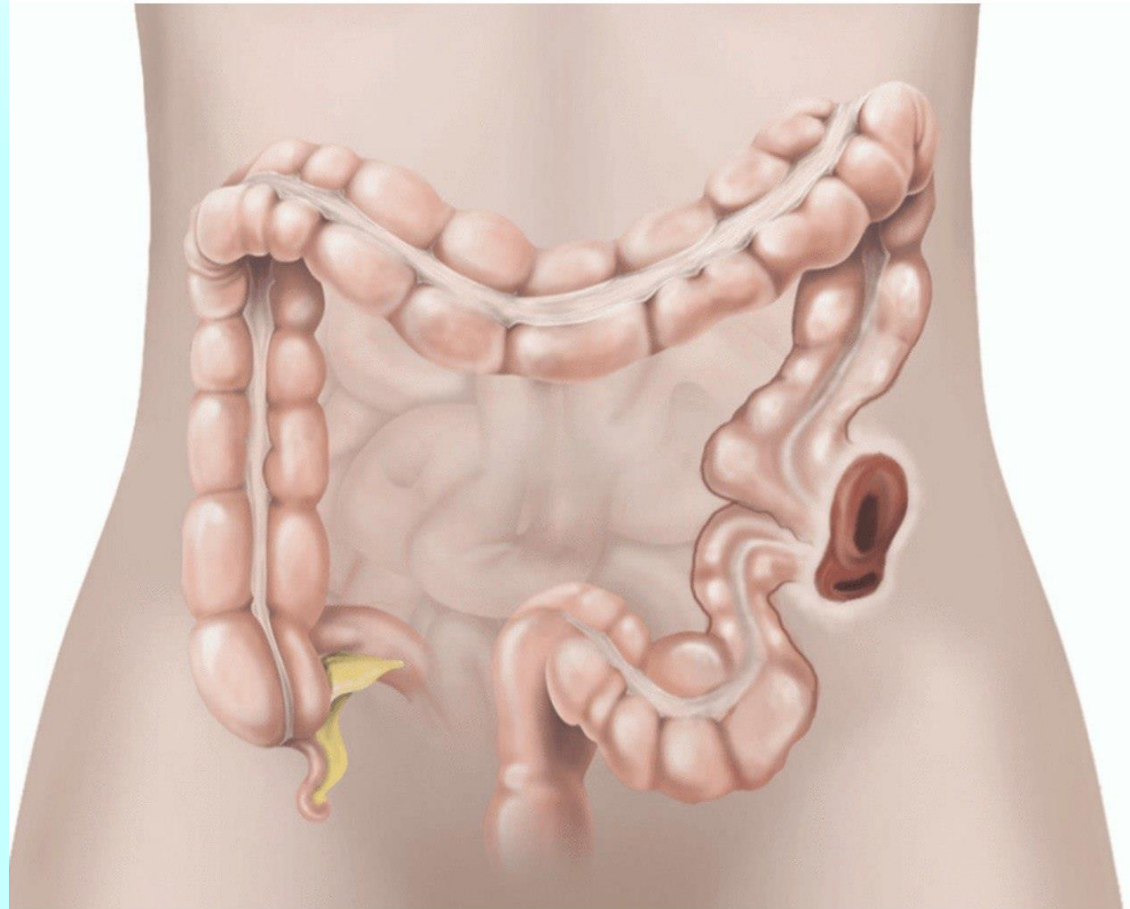
* Amputation because of trauma or disease:

- *Finger amputation.*
- *Lower limb amputation.*

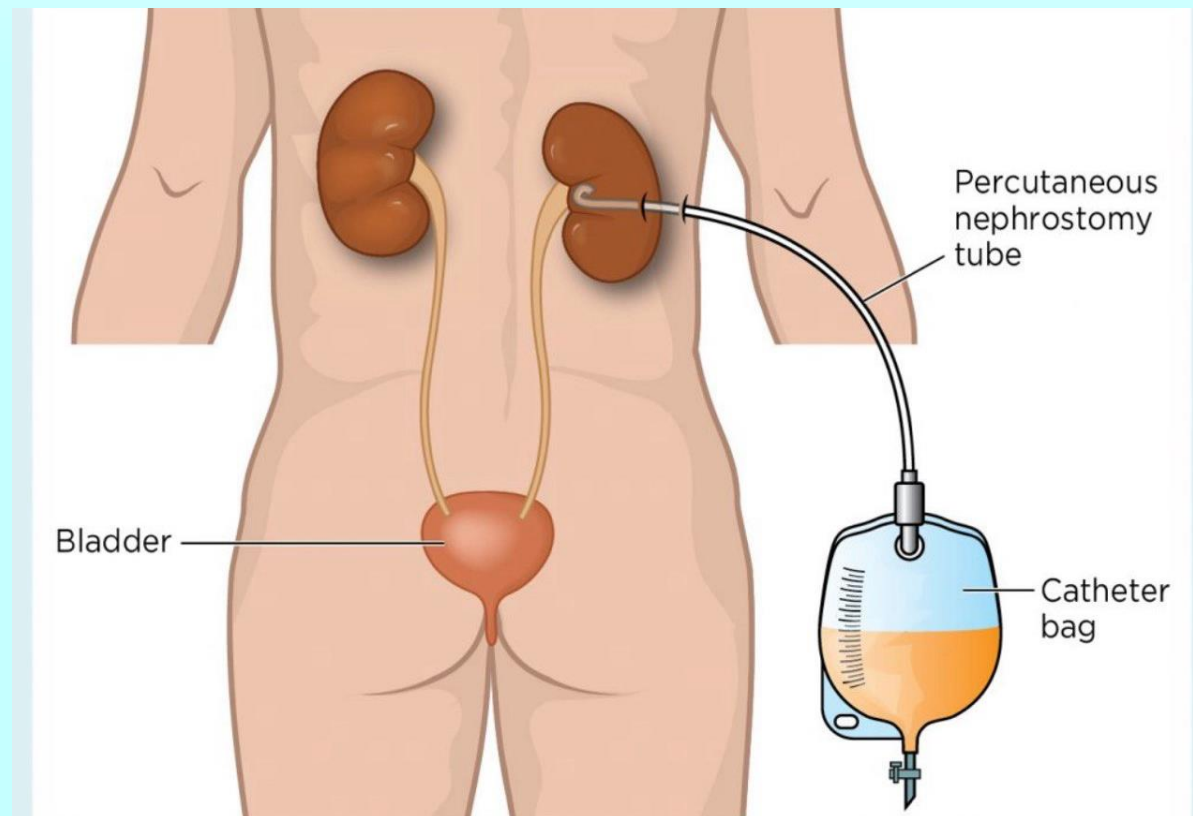
* Skin Graft: *patch of skin that is removed by surgery from one area of the body & transplanted to another area.*

- * **Draining of accumulated fluids (e.g. pus, blood etc.).**
- * **Removal of stones from the urinary system or biliary system.**
- * **Repair of a fistula, hernia, or prolapse.**
- * ***Transplant surgery:* the donor organ (taken out of the donor's body) is inserted into the recipient's body.**
- * **Creation of a stoma in the body. (temporary or permanent):**

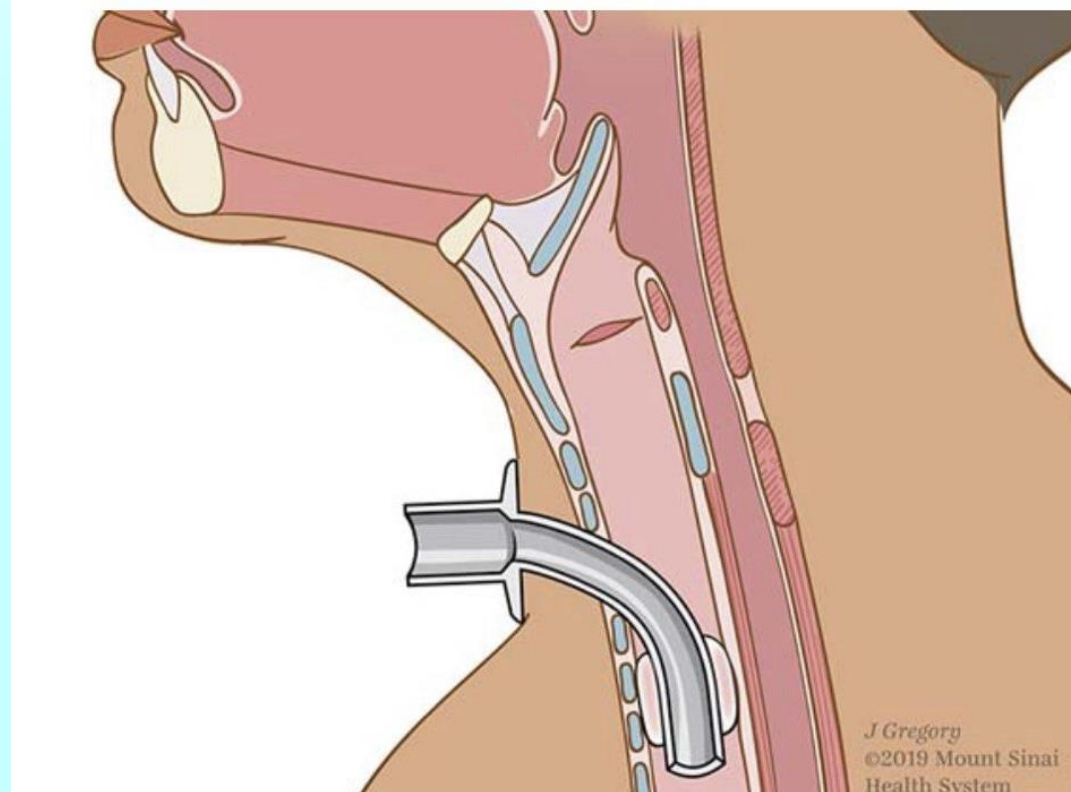
Colostomy: opening (stoma) in the large intestine.



Nephrostomy: opening between the kidney & the skin.



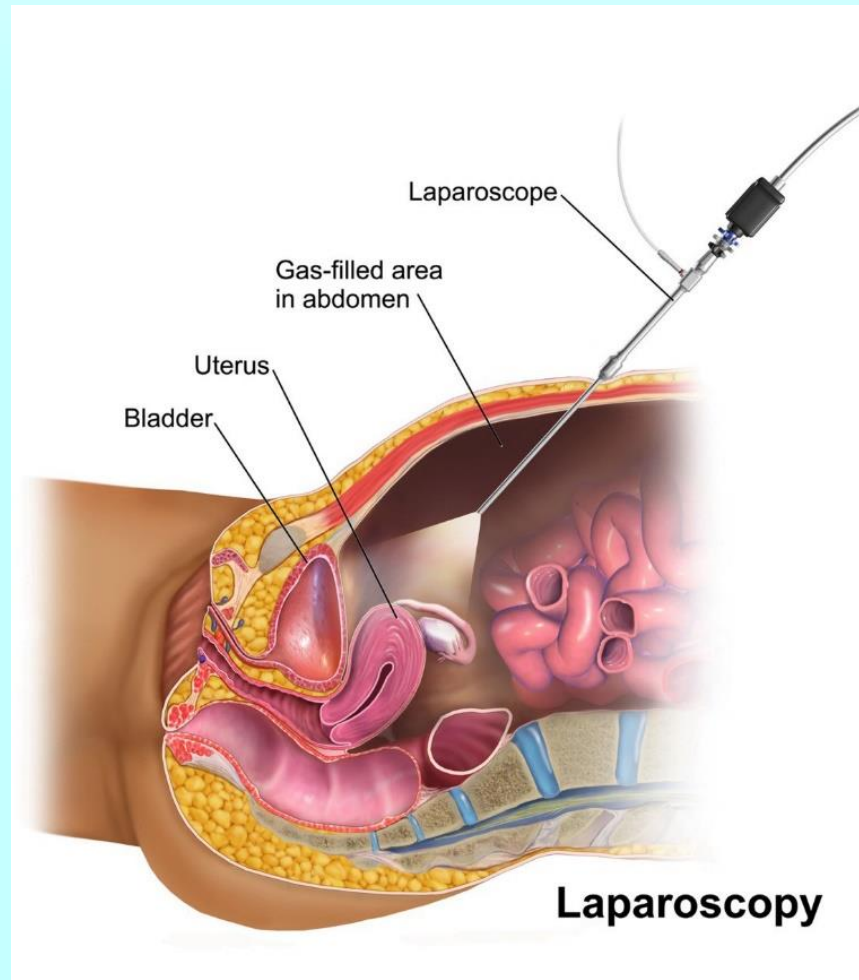
Tracheostomy: opening into the trachea.



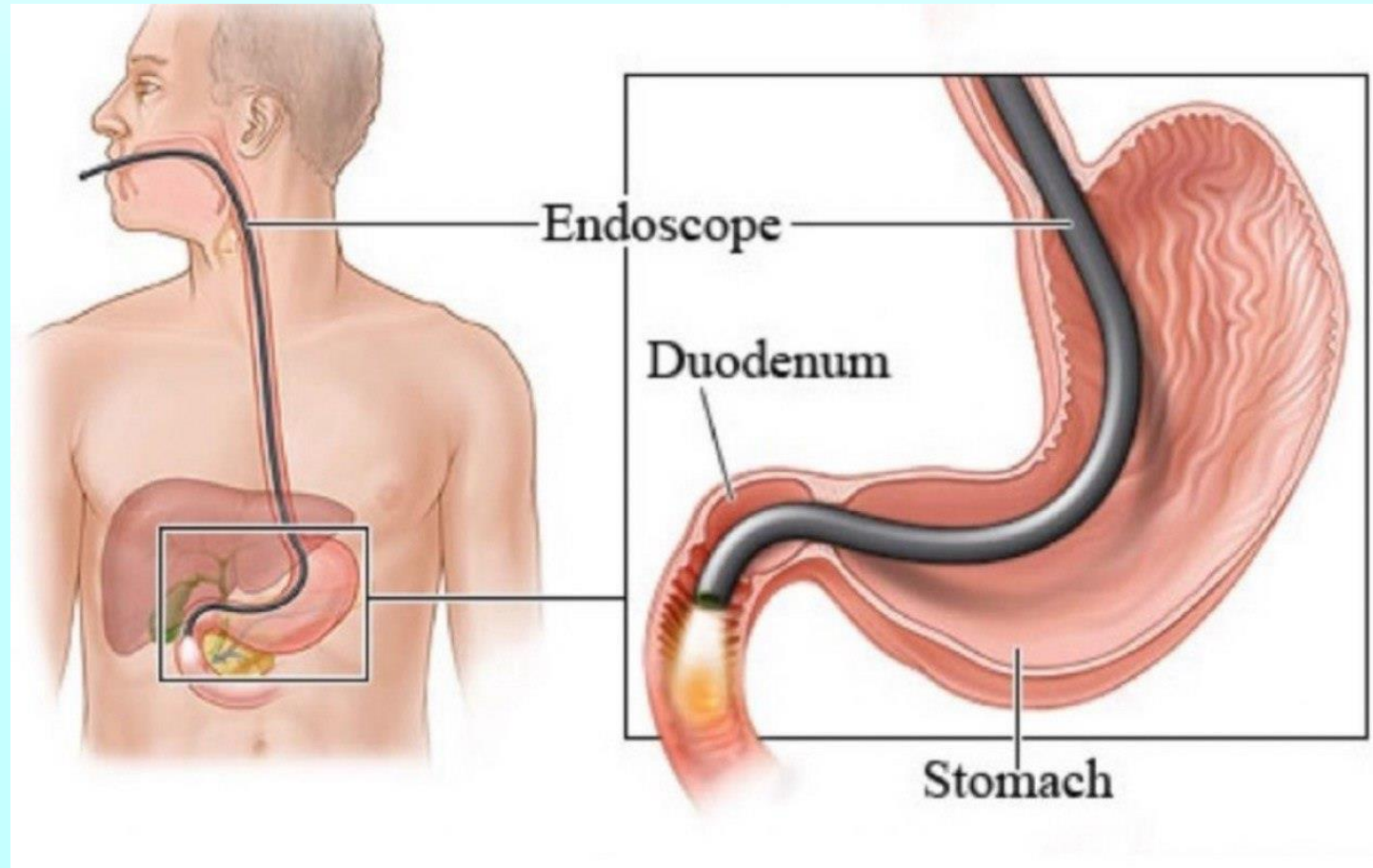
Arthrodesis : uniting of two bones at a joint (joint fusion).



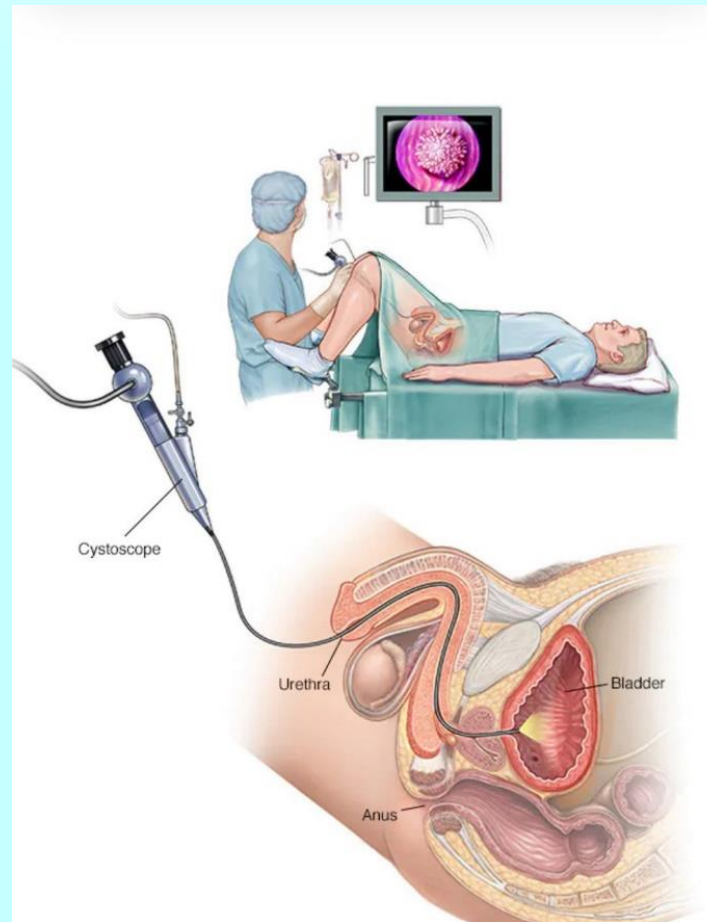
*** Laparoscope: a procedure that uses a laparoscope, inserted through the abdominal wall, to examine the inside of the abdomen & pelvis (for diagnosis & treatment).**



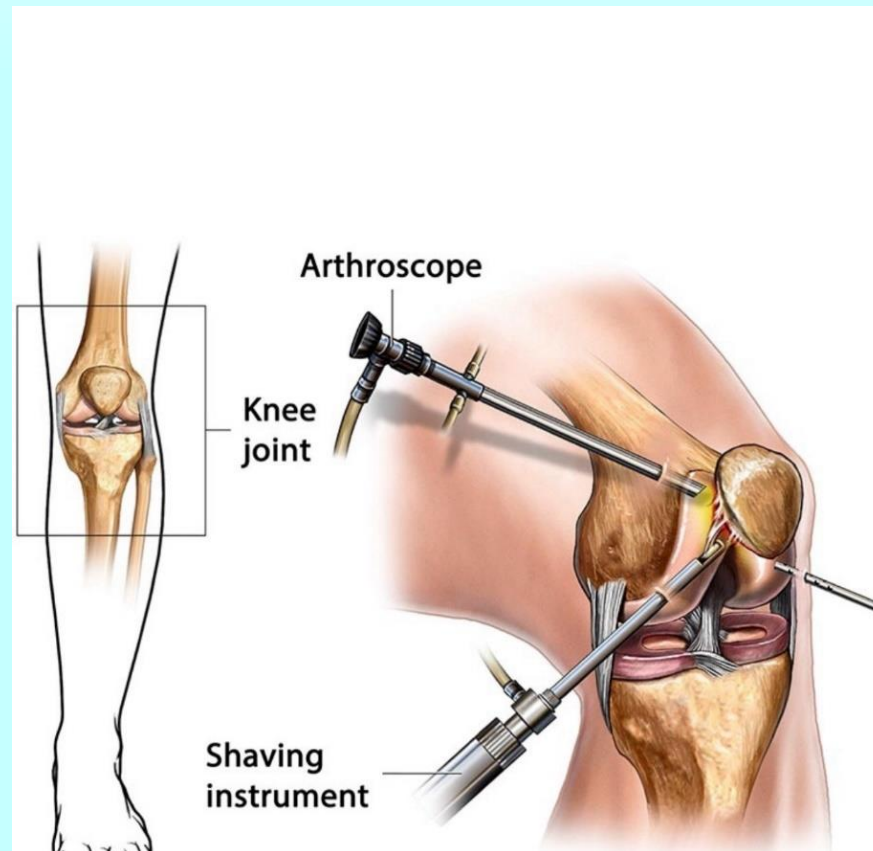
*** Endoscopic procedures:**
OGD (oesophageo – gastro- duodenoscope)



Cystoscope: *surgical procedure to look inside of the bladder & urethra.*



Arthroscopy : surgical procedure to diagnose & treat joints problems



3- Postoperative Care

- ❖ After completion of surgery, the patient is transferred to the post anesthesia care unit & closely monitored.
- ❖ When the patient is recovered from the anesthesia, he is transferred to a surgical ward.
- ❖ During the post-operative period, the patient's general function is assessed, outcome of the procedure is assessed & surgical site is checked for signs of infection or other complications.

Thank You